

Case Study: WitleShop Online Retail System

Background

WitleShop (Pty) Ltd is a South African online retail company that sells electronics, clothing, and home appliances directly to customers through a web and mobile platform.

The company is growing fast and wants to build a new database system to manage:

- Customers
- Products
- Orders
- Payments
- Deliveries
- Suppliers

They have hired you to design the database structure for them.

Business Requirements

The system must support the following:

Customer Management

- Customers must register before placing an order.
- The system must store:
 - Full Name
 - Email (unique)
 - Phone Number
 - Date of Registration
 - Delivery Address(es)
- A customer can have **multiple delivery addresses**.
- A customer can place **multiple orders**.

Product Management

Each product must store:

- Product ID
- Product Name
- Description
- Price
- Stock Quantity
- Category
- Supplier

Business Rules:

- Each product belongs to **one category**
- A category can have **many products**
- Each product is supplied by **one supplier**
- A supplier can supply **many products**

Order Management

When a customer places an order:

- An order must contain:
 - Order ID
 - Order Date
 - Order Status (Pending, Shipped, Delivered, Cancelled)
 - Total Amount
- An order must belong to **one customer**
- A customer can place **many orders**
- An order can contain **multiple products**
- A product can appear in **multiple orders**

Payment Management

- Each order must have a payment record.
- A payment includes:
 - Payment ID

- Payment Date
- Payment Method (Card, EFT, PayFast)
- Payment Status
- Amount Paid

Business Rules:

- One order → one payment
- One payment → belongs to one order

Delivery Management

- Each order must have a delivery record.
- A delivery includes:
 - Delivery ID
 - Delivery Date
 - Delivery Status
 - Courier Name
 - Tracking Number

Business Rules:

- One order → one delivery
- Delivery must be linked to one of the customer's registered addresses.

Student Task

Assignment Instructions

Students must:

1. Identify all entities
2. Identify primary keys
3. Identify foreign keys
4. Identify relationships
5. Show cardinality:

- One-to-one (1:1)
- One-to-many (1:M)
- Many-to-many (M:N)

6. Draw a complete ERD including:

- Entities
- Attributes
- Relationships